1. **ipcmk** - make various System V IPC resources, namely shared memory segments, message queues, and semaphore :
   ipcmk -M size: Create a shared memory segment of size bytes
   ipcmk -S number: Create a semaphore array with number of elements.
   (ipcmk -Q : Create a message queue.)

example:
creating shms egment of 4KiB=4*1024 bytes

```
ubuntu@ubuntu:~$ ipcmk -M 4KiB
Shared memory id: 4
```

verifying using ipcs -m command:

```
ubuntu@ubuntu:~$ ipcs -m

------ Shared Memory Segments --------
key        shmid      owner      perms      bytes      nattch     status
0xfb12ef20 4          ubuntu     644        4096       0
```

creating semaphore array of 3 elements

```
ubuntu@ubuntu:~$ ipcmk -S 3
Semaphore id: 1
```

verifying using ipcs -s command:

```
ubuntu@ubuntu:~$ ipcs -s

------ Semaphore Arrays --------
key        semid      owner      perms      nsems
0x43f1bc53 1          ubuntu     644        3
```

2.**lsipc** - shows information on the System V inter-process communication facilities for which the calling process has read access.

Commonly used options:
—id : Show full details on just the one resource element identified by id.
—global: Show system-wide usage and limits of IPC resources.
these two commands may be combined with one of the three resource options: -m, -q or -s , where
-m, Write information about active shared memory segments.

-q, Write information about active message queues.

-s, Write information about active semaphore sets.

Shared memory segment created as above using ipcmk -M 4kiB returned an id of 4.
The following is the output upon running the lsipc command to list the shm segment corresponding to id=4.

```
ubuntu@ubuntu:~$ lsipc -m --id 4
Key:                                0xfb12ef20
ID:                                 4
Owner:                              ubuntu
Permissions:                        rw-r--r--
Creator UID:                        1000
Creator user:                       ubuntu
Creator GID:                        1000
Creator group:                      ubuntu
UID:                                1000
User name:                          ubuntu
GID:                                1000
Group name:                         ubuntu
Last change:                        Tue Feb 10 04:52:37 2026
Segment size:                       4K
Attached processes:                 0
Status:
Creator PID:                        2154
Last user PID:                      0
```

```
ubuntu@ubuntu:~$ lsipc -m --global
KEY ID PERMS OWNER SIZE NATTCH STATUS CTIME CPID LPID COMMAND RESOURCE DESCRIPTION                                    LIMIT USED  USE%
                                                               SHMMNI   Shared memory segments                           4096    1 0.02%
                                                               SHMALL   Shared memory pages                     18446744073692774399    1 0.00%
                                                               SHMMAX   Max size of shared memory segment (bytes)          16E    -     -
                                                               SHMMIN   Min size of shared memory segment (bytes)           1B    -     -
```

3.**ipcrm** - remove certain IPC resources

System V IPC objects are of three types: shared memory, message queues, and semaphores. Deletion of a message queue or semaphore object is **immediate** (regardless of whether any process still holds an IPC identifier for the object). A **shared memory object is only removed after all currently attached processes have detached** (shmdt(2)) the object from their virtual address space.

Following options can be used along with the command-

-a : removes all resources. Should be used carefully!
-m shmid : removes shm segment corresponding to id: shmid
-s semid: removes semaphore with semid: semid
instead of ids, we can also use keys.
example:

removing shm segment with id 4

```
ubuntu@ubuntu:~$ ipcrm -m 4
```

verifying:

```
ubuntu@ubuntu:~$ ipcs -m

------ Shared Memory Segments --------
key        shmid      owner      perms      bytes      nattch     status
```

```
ubuntu@ubuntu:~$ lsipc -m --id 4
lsipc: id 4 not found
```

4. **ipcs** - show information on IPC facilities

We have been using ipcs to verify the above mentioned commands.
available options:
--id id
Show full details on just the one resource element identified by id. This option needs to be combined with one of the three resource

Resource options
-m, --shmems
Write information about active shared memory segments.

-q, --queues
Write information about active message queues.

-s, --semaphores
Write information about active semaphore sets.

-a, --all
Write information about all three resources (default).

```
ubuntu@ubuntu:~$ ipcs -a

------ Message Queues --------
key        msqid      owner      perms      used-bytes   messages

------ Shared Memory Segments --------
key        shmid      owner      perms      bytes      nattch     status

------ Semaphore Arrays --------
key        semid      owner      perms      nsems
0x43f1bc53 1          ubuntu     644        3
```